

Adilet Shynggys

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RESEARCH INTERESTS

Heat transfer and thermodynamics applied to the efficiency and safety of energy systems, with a focus on nuclear power. Particular interest in reactor thermal-hydraulics — coolant behaviour, boiling, and thermal margins against subcooled boiling, departure from nucleate boiling, and critical heat flux — and in reduced-order models that make plant-scale efficiency and safety tractable. Current experimental work on capillary-driven transport is one route among several toward the same end: getting more useful power out of a thermal system while knowing how far it sits from its limits.

RESEARCH EXPERIENCE

Capillary-Driven Surface Engineering for Power-Plant Efficiency

August 2026 – Present

Department of Mechanical and Aerospace Engineering, Nazarbayev University • Advisor: Prof. Amgad Salama

Improving heat-exchanger and combustion-chamber efficiency in power plants by engineering capillary-driven surface structures that passively transport liquid slugs and droplets along heat-exchanger pipes, reducing flow maldistribution and heat loss without external pumping.

- Fabricated tapered SU-8 microchannels by photolithography on silicon, with width gradients on the order of μm : the taper creates a Laplace-pressure asymmetry intended to drive wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- Running initial transport experiments across a range of taper geometries; quantitative analysis of droplet and liquid-slug velocity, and of how wettability governs transport behaviour, is in progress.
- Relating this geometry-driven transport to liquid supply and rewetting in heated channels, where how quickly coolant reaches a hot surface governs both heat-transfer performance and the margin before dryout.

Micro-Energy-Harvesting Fabrication and Characterization

January – December 2025

Energy Conversion Laboratory, Nazarbayev University

Applying the physical and surface properties of low-cost, simple materials toward sustainable, micro-scale green energy generation via triboelectric charge transfer.

- Fabricated triboelectric nanogenerators for self-powered wearable sensing (e.g., robotic-hand and haptic feedback systems, hiking/outdoor gear) via electrospinning and silicone casting, raising output power density by 15–20%.
- Characterized performance across humidity and cyclic durability testing; benchmarked maximum power output per unit surface area, peak open-circuit voltage, and short-circuit current across candidate applications.
- Built the electrical characterization workflow (load sweeps, data acquisition) in OriginPro and MATLAB, contributing the resulting data and analysis to a co-authored article in *Small* and an award-winning conference poster.

EDUCATION

Nazarbayev University

B.Eng. in Mechanical and Aerospace Engineering

Astana, Kazakhstan

Expected June 2027

- GPA: 3.86 / 4.00 — ranked 1st in cohort; Dean's List, four consecutive semesters.
- Relevant coursework: Fluid Mechanics; Engineering Thermodynamics; Numerical Methods; Differential Equations and Linear Algebra; Engineering Dynamics I–II; Structural Mechanics; Materials and Manufacturing; CAD. In progress: Heat and Mass Transfer; Mechanical Systems Design; Aerospace Propulsion.

The University of Hong Kong

Visiting Exchange Student, Department of Mechanical Engineering

Hong Kong SAR

Spring 2026

- Coursework in Energy Conversion Systems, Design and Manufacture, and Additive Manufacturing; completed two design projects during the exchange (see Engineering Design Projects) and collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

PUBLICATIONS AND CONFERENCE PRESENTATIONS

- [1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al. “Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator,” *Small*, 2026, art. e74486. <https://doi.org/10.1002/smll.74486>
- [2] **A. Shynggys**, S. Yavari, G. Kalimuldina. “Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors.” Presented as a poster at the Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster Award**), and as an abstract at the 13th International Conference on Nanomaterials and Advanced Energy Storage Systems (INESS-2025), Nazarbayev University, Astana, Aug. 2025.

ENGINEERING DESIGN PROJECTS

Reduced-Order Design and Analysis of a VVER-1200 Reactor Core

August 2026 – Present

Senior Capstone Project, Nazarbayev University • Advisor: Prof. Amgad Salama • Team of 4 students, 1 supervisor

- Designing a VVER-1200 core from scratch as part of a four-student team: converting the required electric power output to a core thermal power of 3,200 MWth via cycle efficiency, then using a target fuel burnup near 50 GWd per tonne heavy metal to size fuel pellets, rods, and bundles, arriving at a hexagonal core configuration.
- Preparing thermal-hydraulic calculations for the design point — coolant temperature rise, critical heat flux, and margins to subcooled boiling and departure from nucleate boiling — alongside fuel element, assembly, and core geometry.
- Developing physics-based reduced-order models and thermal-hydraulic sub-models — multiplication factor, power distribution, pressure drop, and heat transfer — each feeding the other, and validating against published reference data to enable rapid parametric design studies and preliminary safety assessment.

Other Engineering Design Projects

2025 – Spring 2026

- *Ergonomic Train-Driver Seat Design* (Mechatronics Engineering Case Competition, sponsored by Wabtec Corporation; team of 3, 1st Place): designed and structurally analyzed a driver's seat for 14-hour locomotive shifts, certifying it against GOST 33330-2015 under static/dynamic fatigue loading.
- *Multi-Stage Transmission (Gearbox) Design* (University of Hong Kong, Prof. Wei Yan): sized shafts using the ASME combined bending–torsion criterion and selected gear modules, keyways, and bearings to meet the project's specified safety margin and fatigue life.
- *Interactive Public Art Installation* (University of Hong Kong, Prof. Homayoon Kazerooni, UC Berkeley): modeled flower-petal forms in SolidWorks to specified dimensional requirements, arranging multiple petals to avoid geometric collisions and verifying structural stability, then 3D-printed each piece on a Bambu Lab A1.

INDUSTRY EXPERIENCE

Product Support Engineering Intern

June – July 2025

Wabtec Corporation — Kazakhstan

- Analyzed thermal performance of locomotive traction motors and supported root-cause review of field temperature excursions.
- Audited shop-floor manufacturing against engineering drawings with a fellow intern, reconciling GOST (shop) with ASME (Wabtec's U.S. standard).
- Traced an insulation-usage inconsistency across units to climate differences between export markets (e.g., uninsulated units for India vs. insulated units for Kazakhstan's winters).

AWARDS AND HONORS

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| • Presidential Scholarship of the Republic of Kazakhstan — national merit scholarship | 2026–2027 |
| • NU President's Scholarship — university merit scholarship | 2024–2025 |
| • Yessenov Foundation Scholarship — top 20 of 500+ applicants | 2026–2027 |
| • Best Research Poster, Belt and Road Initiative Conference | 2025 |
| • 1st Place, Mechatronics Engineering Case Competition (sponsored by Wabtec) | 2025 |

TECHNICAL SKILLS

Programming	MATLAB, Python, Mathematica, $\text{\LaTeX}$
Modeling & Numerics	Coupled neutronic–thermal-hydraulic reduced-order modeling of reactor cores; CHF and DNB margin calculation; validation against published reference data
Simulation & CAD	SolidWorks (CSWA certified), OriginPro
Experimental	Photolithographic microfabrication, electrospinning, silicone casting, 3D printing, electrical device characterization
Design Codes	ASME shaft design, GOST 33330-2015
Languages	Kazakh (native), Russian (fluent), English (IELTS Overall Band 8.0)

TEACHING, LEADERSHIP, AND SERVICE

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| • Physics Tutor, Nazarbayev University — 20+ first-year students. | 2023–2024 |
| • Cohort Representative, Mechanical and Aerospace Engineering. | 2024–2026 |
| • Organizer, Charity Trail Run — 150 runners, 600,000 KZT (~\$1,300) raised. | 2025 |
| • Volunteer, HKU Muslim Society and Mereke Charity Foundation. | 2025–2026 |